

QUCHAN, Iran, Islamic Republic of

WMO: 407400

Lat: 37.117N

Lon: 58.468E

Elev: 1287

StdP: 86.79

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-10.4	-17.0	1.0	-11.5	-13.7	1.3	-7.5	10.2	1.7	9.2	1.0	1.2	60	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	34.5	17.1	33.1	16.6	31.7	16.1	19.5	28.9	18.6	28.3	17.8	27.8	4.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	13.8	23.6	15.3	12.8	23.0	14.3	11.9	22.1	61.8	28.7	58.4	28.4	55.5	27.6	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.5	9.5	8.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-16.5	37.4	4.3	1.6	-19.6	38.6	-22.1	39.5	-24.6	40.4	-27.7	41.6
			-17.4	21.2	4.7	1.0	-20.8	21.9	-23.5	22.5	-26.2	23.1	-29.6	23.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-0.2	2.3	7.7	12.8	17.9	22.3	24.3	22.9	19.5	13.6	6.7	2.2	(6)
(7)		DBStd	9.26	5.07	4.95	4.84	4.38	3.13	2.43	2.30	2.39	2.94	3.81	4.28	4.03	(7)
(8)		HDD10.0	1028	318	218	102	23	0	0	0	0	12	114	242		(8)
(9)		HDD18.3	2599	576	449	331	172	46	2	0	1	21	153	349	499	(9)
(10)		CDD10.0	2021	1	2	30	105	246	369	444	401	286	122	16	1	(10)
(11)		CDD18.3	551	0	0	1	4	33	121	186	143	57	5	0	0	(11)
(12)		CDH23.3	6840	0	1	11	81	407	1491	2267	1758	704	119	3	0	(12)
(13)		CDH26.7	2793	0	0	1	13	104	621	1068	741	225	21	0	0	(13)
(14)	Wind	WSAvg	3.0	2.1	2.7	3.3	3.5	3.4	3.8	3.5	3.0	3.1	2.9	2.6	2.2	(14)
(15)	Precipitation	PrecAvg	288	34	36	54	43	34	14	4	2	4	11	30	25	(15)
(16)		PrecMax	428	78	69	144	105	76	54	16	8	20	46	118	59	(16)
(17)		PrecMin	108	7	2	4	0	0	0	0	0	0	0	0	1	(17)
(18)		PrecStd	70	17	17	26	28	20	15	5	2	6	10	24	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	15.7	17.9	24.8	28.6	31.6	35.3	36.7	35.8	33.5	28.8	22.9	16.0	(19)	
(20)		MCWB	7.1	8.4	11.3	14.1	16.1	17.5	18.0	16.9	15.4	13.8	11.4	7.4	(20)	
(21)		DB	12.4	14.1	21.2	25.6	29.3	33.4	34.8	33.7	31.0	26.4	18.8	13.4	(21)	
(22)		MCWB	5.9	7.2	10.3	13.5	15.9	16.5	17.6	16.4	14.8	13.3	9.7	6.5	(22)	
(23)		DB	9.6	11.9	18.8	23.4	27.5	31.8	33.4	32.0	29.2	24.2	16.6	11.2	(23)	
(24)		MCWB	4.7	6.2	9.7	12.6	15.3	16.1	17.2	15.8	14.6	12.5	8.7	5.8	(24)	
(25)		DB	7.2	9.6	16.1	20.8	25.5	30.3	31.8	30.6	27.4	21.9	14.4	9.0	(25)	
(26)		MCWB	3.4	5.1	8.8	11.9	14.7	15.8	16.8	15.3	14.2	11.7	7.9	4.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	8.0	9.7	12.8	15.9	18.2	19.5	21.1	20.7	18.2	15.2	12.1	8.6	(27)	
(28)		MCDWB	13.0	15.7	20.9	24.3	26.8	30.3	30.7	28.8	26.8	25.3	21.0	14.0	(28)	
(29)		WB	6.4	7.9	11.3	14.4	17.2	18.5	19.9	19.1	17.0	14.0	10.7	7.2	(29)	
(30)		MCDWB	11.4	12.6	18.7	22.8	25.9	29.1	29.6	27.6	26.2	23.7	17.2	11.9	(30)	
(31)		WB	5.0	6.6	10.2	13.5	16.4	17.6	19.0	18.0	16.0	13.2	9.5	6.1	(31)	
(32)		MCDWB	9.4	10.7	17.2	21.2	24.9	28.1	28.8	27.2	25.7	22.2	15.1	10.4	(32)	
(33)		WB	3.5	5.4	9.2	12.5	15.5	16.9	18.1	16.8	15.2	12.3	8.4	4.9	(33)	
(34)		MCDWB	6.9	9.3	15.6	19.2	23.7	27.1	28.3	27.1	25.0	20.3	13.3	8.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.7	10.5	11.8	12.0	13.5	15.0	14.7	15.5	15.7	14.5	11.8	10.6	(35)	
(36)		MCDBR	13.7	13.9	16.1	15.8	16.6	17.9	16.7	16.9	17.7	17.7	15.6	14.4	(36)	
(37)		MCWBR	8.3	7.8	7.3	6.1	6.2	6.6	6.7	7.1	7.5	8.1	8.5	8.6	(37)	
(38)		MCDBR	12.7	11.8	14.3	13.6	14.7	14.9	14.1	13.6	15.1	15.7	13.0	12.6	(38)	
(39)		MCWBR	8.0	7.1	7.2	5.9	6.2	5.9	6.2	6.4	7.0	7.5	7.4	7.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.353	0.429	0.476	0.492	0.472	0.472	0.424	0.390	0.402	0.355	0.312	(40)	
(41)		taud	2.305	2.177	1.968	1.900	1.894	1.950	1.954	2.078	2.150	2.082	2.196	2.321	(41)	
(42)		Ebn at Noon	864	866	825	809	802	817	813	846	855	797	799	839	(42)	
(43)		Edh at Noon	96	124	169	192	196	185	183	158	139	135	107	89	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.30	3.09	4.18	5.51	6.71	7.92	7.85	7.35	6.03	4.27	2.83	2.14	(44)	
(45)		RadStd	0.22	0.35	0.33	0.53	0.36	0.26	0.20	0.29	0.19	0.23	0.25	0.25	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+0.51	-0.82	-0.80	N/A	N/A	N/A	+85	(47)

Nomenclature: See separate page